KAIZEN MATHS

Lesson starter

HCF, LCM and Prime Factorisation

Word Problems

Name _____

S1

How many teeth pass before they align again?

S2

What is the longest equal length that cuts both ropes exactly?

S3

What is the side length of the smallest square they can tile?

Show your working

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NUM-A60B-L4 · S

Show your working

KAIZEN MATHS

Exit ticket

HCF, LCM and Prime Factorisation

Word Problems

Name _____

E1

What is the longest equal length that cuts both ropes exactly?

E2

What is the side length of the smallest square they can tile?

KAIZEN MATHS

Exit ticket

HCF, LCM and Prime Factorisation

Word Problems

Name _____

E1

What is the longest equal length that cuts both ropes exactly?

E2

What is the side length of the smallest square they can tile?

NUM-A60B-L4 · E

Show your working

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Exit ticket

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Name _____

E1

What is the longest equal length that cuts both ropes exactly?

E2

What is the side length of the smallest square they can tile?

NUM-A60B-L4 · E

Show your working

Teacher answers & working

HCF, LCM and Prime Factorisation · Word Problems

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S1

How many teeth pass before they align again?

Answer:

48

Working:

Problem: Two gears have 16 and 24 teeth.

Question: How many teeth pass before they align again?

Solution approach: Find the LCM of the tooth counts.

Numbers: 16 and 24

Prime factorisation of 16: 2^4

Prime factorisation of 24: $2^3 \times 3$

Use the highest power of every prime factor: $2^4 \times 3$

$\mathrm{LCM}(16, 24) = 48$

Answer: 48

S2 / E1

What is the longest equal length that cuts both ropes exactly?

Answer:

2

Working:

Problem: Two ropes are 38 m and 18 m long.

Question: What is the longest equal length that cuts both ropes exactly?

Solution approach: Find the HCF of the two lengths.

Numbers: 38 and 18

Prime factorisation of 38: 2×19

Prime factorisation of 18: 2×3^2

Use the common prime factors: 2

$\mathrm{HCF}(38, 18) = 2$

Answer: 2

Teacher answers & working

HCF, LCM and Prime Factorisation · Word Problems

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S3 / E2

What is the side length of the smallest square they can tile?

Answer:

240

Working:

Problem: Rectangular tiles measure 48 cm by 30 cm.

Question: What is the side length of the smallest square they can tile?

Solution approach: Find the LCM of the two side lengths.

Numbers: 48 and 30

Prime factorisation of 48: $2^4 \times 3$

Prime factorisation of 30: $2 \times 3 \times 5$

Use the highest power of every prime factor: $2^4 \times 3 \times 5$

$\mathrm{LCM}(48, 30) = 240$

Answer: 240